



Nusselt number and development length correlations for laminar flows of water and air in microchannels

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ABSTRACT

Computational Fluid Dynamics (CFD) analyses are carried out to investigate convective heat transfer of developing laminar flows of water and air in microchannels, for wide ranges of parameters ($\alpha = 1-10$, $D_h = 145-375 \mu\text{m}$, $L/D_h = 130-750$, $Q = 1-18 \text{ W}$, $Pr_{in} = 0.7-11.2$, and $Re_{in} = 130-900$). The convergence of the results validated the numerical approach and the mesh grid refinement used and the accuracy is confirmed by the excellent comparison to reported experimental results. The results account for the azimuthal variations of the wall heat flux and temperature, and the changes of the fluid properties with temperature along the microchannel. The compiled database from the present analyses results are used to correlate the local $Nu_{ch}(z)$, which is integrated along the microchannel length to obtain the average Nusselt number, $\bar{Nu}_{ch}$. A continuous correlation of $\bar{Nu}_{ch}$ is developed in terms of Nusselt number for fully developed flow Nu_{fd} , Re_{in} , α , and Pr_{in} , as:

$$\bar{Nu}_{ch} = Nu_{fd} \left[1.0 + 0.0363 / L^{*0.631} \right],$$

$$Nu_{fd} = 8.235 - \left(17.2 / \left(3.785 + (\alpha - 1)^{1.11} \right) \right), \text{ and}$$

$$L^* = \left[(L/D_h) (\alpha^{0.583} / (Re_{in} Pr_{in})) \right]^{1.2}$$

The values of the thermal development length, l_{th} , for $Nu_{ch}(z)$ are also correlated in terms of D_h , α , Re_{in} , and Pr_{in} , as:

$$l_{th} = (0.028 + 0.088e^{-0.45(\alpha-1)}) D_h Re_{in} Pr_{in}$$

The correlations of $Nu_{ch}(z)$, $\bar{Nu}_{ch}$ and l_{th} are in excellent agreement with the present numerical results and those reported by other investigators for $Nu_{ch}(z)$ and $\bar{Nu}_{ch}$. The numerical values of $\bar{Nu}_{ch}$ are 10–20% higher than those reported experimentally, assuming uniform wall and fluid temperatures along the microchannel, and neglecting the effects of the flow development and the azimuthal variations in the wall heat flux and temperature.

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1. Introduction

Microchannels heat sinks are being developed to handle the high thermal power dissipation by compact electronic packages, high power computer chips, and central processing units [1–19].

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Nomenclature

a	microchannel half width (μm)	u	fluid average velocity (m/s)
A	cross section flow area (mm^2)	V	fluid velocity vector (m/s)
b	microchannel half height (μm)	z^*	dimensionless length, $[(z/D_h)(\alpha^{0.583}/Re_{in}Pr_{in})]^{1.2}$
CFD	Computational Fluid Dynamics	x, y, z	Cartesian coordinates
C_p	heat capacity (J/K)	x^*	dimensionless length, $L/(D_h Re_{in} Pr_{in})$
Cu	copper substrate		
D_h	equivalent hydraulic diameter (μm)	Subscript	
h	heat transfer coefficient ($\text{W/m}^2 \text{K}$)	b	bulk, microchannel bottom wall
j	operator, j = b, bottom; j = s, sides; and j = t, top	ch	microchannel
k	thermal conductivity (W/m K)	in	microchannel inlet, interface
L	microchannel length (mm)	f	fluid
L^*	dimensionless microchannel length, $[(L/D_h)(\alpha^{0.583}/(Re_{in}Pr_{in}))]^{1.2}$	fd	fully developed
L_{th}	dimensionless thermal development length, $l_{th}/D_h Pr_{in}$	th	thermal
Nu	Nusselt number, $h D_h/k$ (–)	s	solid
$Nu_{ch}(z)$	microchannel local Nusselt number, $h(z) D_h/k$, Eq. (8)	w	microchannel inner wall
$\bar{Nu}_{ch}$	Microchannel average Nusselt number, Eq. (10)		
P	Pressure (Pa)	Greeks	
p	microchannel pitch (mm)	ΔT	temperature difference (K)
Pr	Prandtl number, $\mu C_p/k$, (–)	α	microchannel aspect ratio, b/a
Q	thermal power (W)	μ	dynamic viscosity (Pa·s)
Re	Reynolds number, $\rho D_h u/\mu$, (–)	ρ	density (kg/m^3)
T	temperature (K)	l_{th}	thermal development length (μm)

The microchannels heat sinks are typically cooled by developing laminar flow of water. The miniaturized dimensions of the microchannels significantly increase the contact area for heat transfer, but at the expense of experiencing large friction pressure losses. For hydraulic diameters, D_h , less than $10^3 \mu\text{m}$, the apparent slip developing at the walls could markedly decrease the friction number below that of the theoretical Hagen-Poiseuille with no slip [19–25]. The decrease in the friction number for microchannels with $D_h < 10^3 \mu\text{m}$ could be as much as 60%, depending on the inlet Reynolds number, Re_{in} , the apparent slip length, β , and the aspect ratio of the microchannel, α [25].

These parameters as well as the inlet temperature, the wall heat flux, and the entrance effect in the microchannels, affect the local and the average Nusselt numbers, and hence the performance of the heat sinks. For the 3-D conjugate heat transfer in microchannels, the wall temperature and heat flow vary azimuthally and so is the local Nusselt number, $Nu_{ch}(z)$. The azimuthal variations in the heat flow and wall temperature in the microchannels could not be measured experimentally, but can accurately be determined numerically from the solution of the 3-D conjugate heat transfer problem. The experimentally reported values of $\bar{Nu}_{ch}$, based on the average wall and fluid temperatures, neglect the azimuthal variations of the wall temperature and heat flux, and the flow development in the microchannel. The values of $Nu_{ch}(z)$ are highest at the entrance of the microchannels due to the flow mixing and decrease gradually with distance from the entrance till reaching a constant value, Nu_{fd} , when the flow becomes fully developed. The flow development length of Nusselt number depends not only on the microchannel's aspect ratio, α , and D_h , but also on Re_{in} and Pr_{in} . Thus, depending on the values of the flow development length, l_{th} , the actual Nusselt number, $\bar{Nu}_{ch}$, would be higher than experimentally reported values. Therefore, it is desirable to quantify the differences and parametrically investigate their dependence on the microchannel dimensions (α , D_h , L) and inlet flow condition (Re_{in} and Pr_{in}).

Numerous numerical and experimental studies of convection heat transfer in microchannels have been reported during the last four decades. Many had investigated the effects of α and D_h , on

convection heat transfer of uniform inlet velocity flows in microchannels [2,3,5,6,13,14]. A few had investigated the effects of using internal fins, grooves, or wavy shaped microchannels on the pressure losses and the convection heat transfer coefficient [8–10,26–28]. In the experiments, the average wall temperature and the flow bulk temperature are used to determine the microchannel average Nusselt number, $\bar{Nu}_{ch}$, assuming constant fluid properties. The fluid properties are evaluated either at the flow inlet or bulk temperature in the microchannel.

On the other hand, numerical analyses of the 3-D conjugate heat transfer in microchannels could account for the effect of the azimuthal variations of the wall temperature and heat flux on the local values and the axial distribution of $Nu_{ch}(z)$ along the microchannel length. Some numerical solutions assumed constant fluid properties [9,11,13,14,18,20], others accounted for the changes in the fluid properties with temperature along the microchannel length [12,26], and some neglected conduction in the microchannel wall and focused solely on the flow, assuming a constant wall heat flux [15–17]. The values of $\bar{Nu}_{ch}$ are then determined from the integration of the obtained distribution of $Nu_{ch}(z)$ along the length of the microchannel. Many correlations for $\bar{Nu}_{ch}$ have been reported based on the results of numerical analyses and experimental measurements of the average wall and fluid temperatures. Fig. 1 shows large variances in the predictions of these correlations, some of which are limited to a narrow range of parameters [5,6,15,16,18,27].

Kim [2] experimentally investigated convection heat transfer of laminar water flow in 10 aluminum-alloy rectangular microchannels ($\alpha = 0.25$ – 3.8 and $D_h = 154.9$ – $580.5 \mu\text{m}$) and reported values of $\bar{Nu}_{ch}$ as a function of Re_{in} (100–1200). Wu and Cheng [3] conducted similar experiments for water flow in heat sinks of 13 silicon microchannels with triangular and trapezoidal cross sections. The experimentally determined values of $\bar{Nu}_{ch}$ were correlated as a function of α , D_h , and Re_{in} ; however, there was an average deviation of 20% between the proposed correlation and the obtained experimental values.

Lee et al. [4] experimentally studied convective heat transfer of deionized water flow in 5 rectangular copper microchannels, with

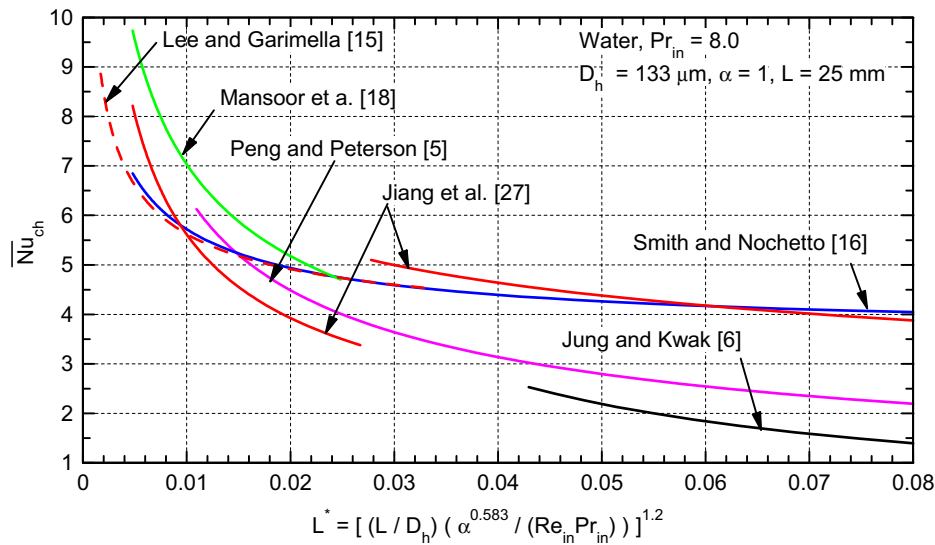


Fig. 1. Comparison of reported $\overline{Nu}_{ch}$ correlation for laminar water flow in microchannels.

$\alpha = 4.56$ – 5.59 and $D_h = 318$ – $903 \mu\text{m}$, and reported values of $\overline{Nu}_{ch}$ as a function of Re_{in} . Peng and Peterson [5] experimentally investigated pressure losses and convective heat transfer of water in stainless-steel microchannels, with $D_h = 133$ – $367 \mu\text{m}$ and $\alpha = 0.5$ – 3.0 , and correlated the values of $\overline{Nu}_{ch}$ as a function of Re_{in} , Pr_{in} , D_h and α (Fig. 1, and Table 1). The mean deviation of the proposed correlation from the values of $\overline{Nu}_{ch}$ in the experiments was 30%.

Jung and Kwak [6] experimentally determined the convective heat transfer coefficients and the friction numbers for laminar water flows in silicon rectangular microchannels, with $\alpha = 0.5$, 0.667 and 1.0 and $D_h = 133$, 120 and $100 \mu\text{m}$. They correlated their experimental values of $\overline{Nu}_{ch}$ in terms of α , Re_{in} and Pr_{in} (Fig. 1 and Table 1); however, the correlation predictions were within 50% the $\overline{Nu}_{ch}$ values in the experiments.

Jiang et al. [27] experimentally studied convection heat transfer of water flows in copper heat sinks of microchannels with $\alpha = 0.5$ – 4 and $D_h = 133$ – $320 \mu\text{m}$. The obtained values of $\overline{Nu}_{ch}$ were correlated as a function of Re_{in} , D_h , and Pr_{in} , and, and the microchannel length, L (Fig. 1 and Table 1). The proposed correlation was within 15% of the values of $\overline{Nu}_{ch}$ in the experiments.

Zhai et al. [7] performed experiments, which investigated convection heat transfer of water flow in a silicon microchannel heat sink, $D_h = 133 \mu\text{m}$ and $\alpha = 2$, and at $Re_{in} = 130$ – 850 . They compared the experimentally determined values of $\overline{Nu}_{ch}$ with the values calculated numerically using a 3-D conjugate model, and proposed a correlation. The numerical values of $\overline{Nu}_{ch}$ were consistently higher than those obtained experimentally by $\sim 10\%$.

Sui et al. [8] numerically investigated pressure losses and convective heat transfer of deionized water flows in straight and sinusoidal copper microchannels, with a rectangular cross section and $D_h = 272 \mu\text{m}$, at $Re_{in} = 300$ – 800 . They calculated $\overline{Nu}_{ch}$ and the friction factor in the straight and wavy microchannels as functions of Re_{in} . The wavy microchannel results showed 211% enhancement in the convective heat transfer coefficient, with 76% more pressure losses than in the straight microchannel.

In their experiments, Wang et al. [9] examined the effect of using micro-scale ribs and grooves on the convective heat transfer of water flows in silicon microchannel, with $\alpha = 1.43$ and $D_h = 412 \mu\text{m}$, at different values of Re_{in} . The determined values of $\overline{Nu}_{ch}$ for the rib-grooved microchannel were higher than those of

the smooth microchannel, but at the expense of incurring higher pressure losses. Natrajan and Christensen [10] conducted experiments to investigate the effect of wall surface roughness on the convective heat transfer of water flows in microchannels with $D_h = 600 \mu\text{m}$ and RMS surface roughness of 0.09 – $15.06 \mu\text{m}$. The determined $Nu_{ch}(z)$ within the thermal developing length, near the entrance of the microchannel, increased with increased surface roughness, while that for fully developed flow, Nu_{fd} , was the same for the rough and smooth microchannels. At $Re_{in} = 1200$, the reported $\overline{Nu}_{ch}$ values in the rough-wall microchannel was 25% higher than in the smooth microchannel.

Qu and Mudawar [11] compared the experimental pressure losses and temperature of the deionized water flow in a copper microchannel, with $D_h = 349 \mu\text{m}$ at $Re_{in} = 139$ – 1600 , to those calculated from the numerical solution of the 3-D Conjugate heat transfer in the microchannel, assuming constant fluid properties. The numerical and experimental values were in a fairly good agreement. Li et al. [12] numerically investigated the effect of treating the water properties on the convection heat transfer of water flow in a rectangular microchannel, with $D_h = 333 \mu\text{m}$ at $Re_{in} = 101$ – 177 . They considered three conditions: (a) assuming water constant properties equal to those at the inlet temperature of 293 K , (b) assuming constant properties equal to those at the flow bulk temperature, and (c) using temperature dependent properties along the microchannel. The latter resulted in lower friction number and higher $Nu_{ch}(z)$ values.

Gunnasegaran et al. [13] performed numerical investigations of convection heat transfer in heat sink of microchannels with triangular, rectangular and trapezoidal cross sections, at $Re_{in} = 100$ – 1000 . The reported values of $\overline{Nu}_{ch}$, and the pressure losses were higher for the rectangular microchannel. Sahar et al. [14] numerically investigated the effects of changing α and D_h on convection heat transfer of water flow in a rectangular microchannel heat sink at $Re_{in} = 500$ – 2000 . Results indicated that $\overline{Nu}_{ch}$ was weakly dependent on α , but increased with increased D_h .

Lee and Garimella [15] performed numerical analysis of convection heat transfer of hydrodynamically fully developed water flow in rectangular microchannels, with $\alpha = 1$ – 10 , and axially uniform wall heat flux. They correlated their results of the thermal development length, ι_{th} , $Nu_{ch}(z)$, and $\overline{Nu}_{ch}$. The proposed $\overline{Nu}_{ch}$ correlation (Fig. 1, and Table 1), as based on the results of assuming constant water thermal properties and neglecting the conjugate heat

Table 1
Reported experimental and numerical correlations of $\overline{Nu}_{ch}$ in microchannels.

Reference	Method	Setup	Re _{in}	Reported correlation
Peng et al. [5]	Exp.	Microchannels (Stainless Steel) 133 < D _h (μm) < 367 0.5 < α < 3.0 2.0 < p < 4.6 3.9 < Pr _{in} < 7.0 Water	<1000	$\overline{Nu}_{ch} = 0.1165(D_h/p)^{0.81} \alpha^{0.79} Re_{in}^{0.62} Pr_{in}^{1/3}$
Jung and Kwak [6]	Exp.	Microchannels (Silicon) 100 < D _h (μm) < 134 1.0 < α < 2.0 6.3 < Pr _{in} < 6.8 Water	<330	$\overline{Nu}_{ch} = 0.00058(\alpha)^{-0.3} Re_{in}^{1.15} Pr_{in}^{1/3} (\mu_{in}/\mu_w)^{2.76}$
Jiang et al. [27]	Exp.	Microchannels (Copper) 133 < D _h (μm) < 367 1 < α < 10 Water	<600	$\overline{Nu}_{ch} = 0.52(Re_{in} Pr_{in} D_h/L)^{0.62}; (L/D_h Re_{in} Pr_{in}) < 0.05$ $\overline{Nu}_{ch} = 2.02(Re_{in} Pr_{in} D_h/L)^{0.31}; (L/D_h Re_{in} Pr_{in}) > 0.05$
Mansoor et al. [18]	Num.	Microchannels (Copper) D _h = 318 μm α = 4.56 Pr _{in} = 5.83 Water – conjugate heat transfer – constant fluid properties – uniform inlet velocity	<2000	$\overline{Nu}_{ch} = 0.293 Re_{in}^{0.53} Pr_{in}^{1/4}$
Lee and Garimella [15]	Num.	Microchannels 133 < D _h (μm) < 367 1 < α < 10 Pr _{in} = 5.83 Water – uniform azimuthal heat flux – uniform azimuthal temperature – constant fluid properties – hydrodynamically fully developed inlet flow	<1500	$x_{th}^* = -1.275 \times 10^{-6} \alpha^6 + 4.709 \times 10^{-5} \alpha^5 - 6.902 \times 10^{-4} \alpha^4 + 5.0145 \times 10^{-3} \alpha^3$ $-1.769 \times 10^{-2} \alpha^2 + 1.845 \times 10^{-2} \alpha + 5.691 \times 10^{-2}$ $\overline{Nu}_{ch} = \frac{1}{C_1(x^*)^2 + C_3} + C_4; x^* < x_{th}^*$ $C_1 = -2.757 \times 10^{-3} \alpha^3 + 3.274 \times 10^{-2} \alpha^2 - 7.464 \times 10^{-5} \alpha + 4.4764$ $C_2 = 0.6391$ $C_3 = 1.604 \times 10^{-4} \alpha^2 - 2.622 \times 10^{-3} \alpha + 2.568 \times 10^{-2}$ $C_4 = 7.301 - 13.11 \alpha^{-1} + 15.19 \alpha^{-2} - 6.094 \alpha^{-3}$
Smith and Nochetto [16]	Num.	Macro/Micro channels D _h = 333 μm 1 < α < 100 Pr _{in} = 5.83 Water – uniform azimuthal heat flux – uniform azimuthal temperature – constant fluid properties – hydrodynamically fully developed inlet flow	<2000	$\overline{Nu}_{ch} = [(Nu_{ch,x^*=0})^n + (Nu_{fd})^n]^{1/n}$ $Nu_{ch,x^*=0} = 2.053(x^*)^m [1 - 1.016\alpha^{-1} + 1.281\alpha^{-2} - 0.5659\alpha^{-3}]$ $Nu_{fd} = 8.235[1 - 2.0421\alpha^{-1} + 3.0853\alpha^{-2} - 2.4765\alpha^{-3} + 1.0578\alpha^{-4} - 0.1861\alpha^{-5}]$ $m = -0.3302[1 + 0.1083\alpha^{-1} - 0.06569\alpha^{-2}]$ $n = 3.673[1 - 0.3279\alpha^{-1} + 0.2924\alpha^{-2}]$

transfer in the microchannel walls. They also compared their $\overline{Nu}_{ch}$ correlations with the experimental results of Lee et al. [4]. Smith and Nochetto [16] extended the work of Lee and Garimella on convection heat transfer of water flow to microchannels, with α = 1–100, and reported similar correlations for the thermal development length, l_{th} and $\overline{Nu}_{ch}$ (Fig. 1 and Table 1).

McHale and Garimella [17] conducted numerical analysis of convective heat transfer of thermally developing water flows in trapezoidal microchannels, with α = 1–100, assuming axially uniform wall heat flux, circumferentially uniform wall temperature, and constant water properties. They correlated the obtained values of l_{th} , Nu (z), and $\overline{Nu}_{ch}$ in terms of α and Graetz number.

Mansoor et al. [18] numerically investigated convection heat transfer of deionized water flow in a copper microchannel, with D_h = 318 μm, α = 4.56 and total length of 25.4 mm, at Re_{in} = 500–2000. The water properties were assumed constant and equal to those at the inlet temperature. They correlated their numerical results of l_{th} and $\overline{Nu}_{ch}$ in terms of Re_{in} and Pr_{in}. The proposed $\overline{Nu}_{ch}$ correlation was within 5% of the experimentally determined values for the same microchannel.

In summary, the reported experimental and numerical correlations for $\overline{Nu}_{ch}$ in microchannels are not for the same conditions, and hence provide different predictions. The reported experimental data and correlations (Fig. 1 and Table 1) neither account for

the axial variations in the wall and the fluid temperatures nor the entrance effect, thus underestimates $\overline{Nu}_{ch}$. On the other hand, the reported numerical investigations of the conjugate heat transfer in microchannels account for the axial variation in both the wall and fluid temperatures as well as the azimuthal variations in the wall temperature and heat flux. Therefore, the numerical results of the local mussel number, Nu_{ch}(z), are realistic and could be used to determine the flow thermal development length, l_{th} , and the microchannel average Nusselt number, $\overline{Nu}_{ch}$ with reasonable accuracy. However, assuming constant fluid properties, would affect the numerical values of Nu_{ch}(z), and $\overline{Nu}_{ch}$. Although the numerical values of $\overline{Nu}_{ch}$ would be expected to be higher than the experimentally determined values; the difference has not been quantified for a wide range of operating and geometrical conditions, which is a focus of this work.

The comparison of the reported experimental and numerical correlations in Fig. 1 shows a wide variation in the predictions of $\overline{Nu}_{ch}$ in microchannels. The differences range from 34% to as much as 57% at low and high Re_{in} values, respectively. In addition, the reported correlations for predicting the thermal development length in microchannels, l_{th} , are very limited. Therefore, there is a need to conduct comprehensive parametric numerical analyses of the 3-D conjugate heat transfer in microchannels, for wide ranges of geometrical parameters (α, D_h, L), heating rate, Q, and

both Pr_{in} and Re_{in} , and use the results to generate a comprehensive database for correlating $Nu_{ch}(z)$ and $\bar{Nu}_{ch}$, as well as l_{th} . These analyses need to account for the changes in the fluid properties with temperature along the microchannel length, and the azimuthal variations in both the wall heat flux and temperature. This is also a focus of the present work.

In this work, Computational Fluid Dynamic (CFD) analyses investigated conjugate heat transfer in microchannels for a wide range of parameters. The accuracy of the results using the appropriate numerical mesh refinement are validated by comparing the calculated pressure losses and wall temperatures to the reported experiments measurements by Qu and Mudawar [11], for water flow in a copper heat sink with rectangular microchannels, $\alpha = 3.08$, $L/D_h = 128.264$, $D_h = 231 \mu m$, $T_{in} = 288.15 K$, $Re_{in} = 130$ – 1600 , and $Q = 20.21 W$ and $40.45 W$. The validated numerical approach is then used to conduct comprehensive analyses of convection heat transfer in microchannels for wide ranges of geometrical (D_h , L , and α) and operation (Re_{in} and Pr_{in}) parameters and thermally and hydrodynamically developing laminar flows of water and air. The investigated parameters are $\alpha = 1$ – 10 , $D_h = 145$ – $375 \mu m$, $Pr_{in} = 0.7$ – 11.2 , $Re_{in} = 130$ – 900 , $Q = 1$ – $18 W$, and $L/D_h = 30$ – 750 .

The calculated values of the azimuthal wall temperature and heat flux as well as of the flow bulk temperature are used to calculate the local values of Nusselt number, $Nu_{ch}(z)$, along the microchannel. The performed numerical analyses with different values of Q , and a uniform axial heat input, account for the changes in the fluid properties with temperature. To obtain $\bar{Nu}_{ch}$, the calculated values of $Nu_{ch}(z)$ are integrated over the microchannel length. The present numerical values of $Nu_{ch}(z)$ are also used to determine those of the thermal development length, l_{th} , as functions of Re_{in} and Pr_{in} , and correlate them in terms of the operation and geometrical parameters (α , D_h/L , Re_{in} , and Pr_{in}). In addition, a semi-empirical correlation of the present numerical values of $\bar{Nu}_{ch}$ is developed and compared to those reported by others based on both numerical analyses and experimental measurements of the microchannel average wall and fluid temperatures (Table 1 and Fig. 1).

2. Approach

The performed CFD numerical analyses of the conjugate heat transfer in microchannels are for steady state laminar flows of water and air. These analyses, for both fully developed and developing flow conditions and different input thermal powers, Q , are carried out using the STAR-CCM+ commercial software package [29]. They account for the 3-D heat transfer both in the solid and fluid regions and the changes in the fluid properties with temperature. The calculated values of $Nu_{ch}(z)$, along the microchannel account for the azimuthal variations in the local temperatures and heat fluxes at interface between the flowing fluid and the surrounding solid.

The numerical analyses solve the 3-D steady-state continuity and energy balance equations in the fluid, with symmetrical spatial distributions of the flow velocity and temperatures in the microchannel. The governing equations in the fluid region are thermally coupled to the 3-D heat conduction in the surrounding solid. The governing equations in the fluid region are given as:

$$\text{Continuity : } \nabla \cdot (\rho V) = 0, \quad (1a)$$

$$\begin{aligned} \text{Momentum balance : } \nabla \cdot (\rho V V) \\ = -\nabla P + \nabla \cdot \mu (\nabla V + \nabla^T V) + \rho g, \text{ and} \end{aligned} \quad (1b)$$

$$\text{Energy balance : } \nabla \cdot (\rho c_p T V) = \nabla \cdot (k_f \nabla T). \quad (1c)$$

The steady-state 3-D heat conduction equation in the surrounding solid is given as:

$$k_s \nabla^2 T_s = 0 \quad (2)$$

Eqs. (1) and (2) are solved simultaneously, subject to the continuity of the local heat fluxes and temperatures at the fluid-solid azimuthal interfaces, as:

$$-k_s (\nabla T_s)_{int} = -k_f (\nabla T_f)_{int}. \quad (3a)$$

$$(T_s)_{int} = (T_f)_{int}. \quad (3b)$$

The present 3-D numerical analyses first investigated the conjugate heat transfer in one microchannel in the heat sink used in the experiments by Qu and Modawar [11]. Fig. 2 presents cross-sectional views of this microchannel with the surrounding Cu wall, the bottom Cu substrate, and the top solid thermal insulation, as in the experiments [11]. Additional input parameters to the numerical analyses include the total thermal power input, Q , the water flow rate or Re_{in} , the water inlet temperature, T_{in} , or Pr_{in} , and the inlet flow condition (uniform velocity or a hydrodynamically fully developed). The obtained numerical results, of the pressure losses and the wall temperatures along the microchannel, are validated by comparing them to the reported measurements in the experiments [11].

The geometrical parameters for the microchannel numerical analyses are given in Fig. 2 and listed in Table 2. The copper substrate and the top polycarbonate plastic insulation, which extend along the microchannel length, are 5.64 mm and 12.7 mm thick, respectively. The insulation has a thermal conductivity of 0.2 W/m K and is cooled at the top surface by natural circulation of ambient air [11] at 298.15 K (Fig. 2). The input thermal power, Q , is applied uniformly at the bottom surface of copper substrate (Fig. 2). Four thermocouples are spaced axially in the Cu substrate and placed 2.46 mm from microchannel bottom. They are used in the experiments to measure the local temperatures in the Cu substrate (Fig. 2). The measured wall temperatures using these thermocouples in the experiments are compared to the calculated values in the present numerical analyses.

2.1. Numerical mesh grid

An important consideration in the present analyses is to ensure that the numerical results convergence is practically independent of the refinement of the implemented grid of the numerical mesh elements. In order to evaluate the sensitivity of the results to the refinement of the implemented mesh grid, the determined values of $\bar{Nu}_{ch}$ using four different grids, with increasing refinements, namely: coarse, intermediate, fine and finer, are compared. Fig. 3 presents schematics of fine numerical grid, while Table 2 lists the details of the microchannel geometrical parameters and input conditions used in the present numerical analyses, and Tables 3 and 4 compare the details of the implemented mesh grids in the analyses.

The numerical grids implemented in the present CFD analyses (Tables 3 and 4) use hexahedral mesh elements and apply prism layers in the fluid at the interfaces with the microchannel walls, to accurately capture the temperature and velocity gradients in the boundary layer (Eq. (3)). For example, in the fine numerical grid there are 5 prism layers, with a growth multiplier of 1.2, at the fluid-Cu wall interfaces. In addition, the numerical mesh grids in Tables 3 and 4 use 3 prim layers with a growth multiplier of 1.2, at all solid-solid interfaces (Fig. 3).

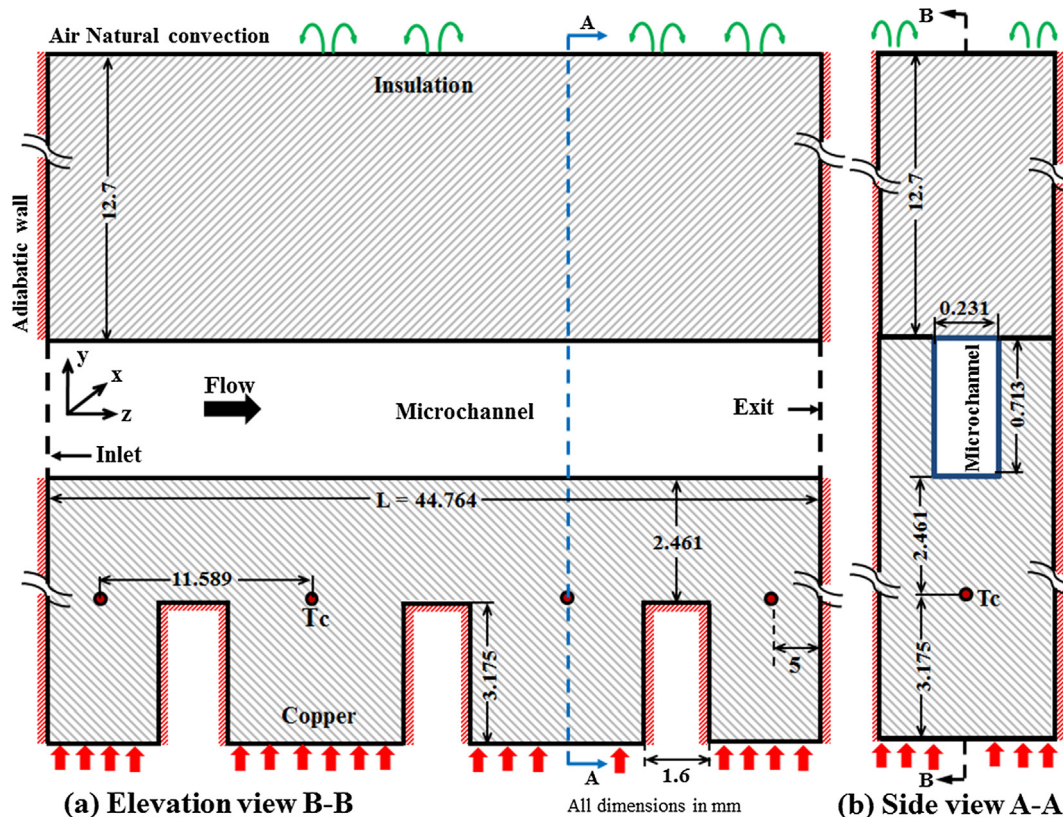


Fig. 2. Cross sectional view of a microchannel setup as in the experiments [11].

Table 2

Geometrical and operation parameters used in the present numerical analyses of convection heat transfer of water flow in a microchannel.

Region	Width (μm)	Height (μm)	Length, L/D_h	D_h (μm)	Fluid	Re_{in}
Fluid flow	231	713	128.264	349	Water or air	130–1200
Cu substrate	467	5640	128.264	–		–
Top insulation	467	12,700	128.264	–		–

As listed in Table 3, the course mesh grid has a total of 0.716 million elements, and 3 prim layers with a growth multiplier of 1.2 in the fluid at the interfaces with the solid Cu and the top insulation (Fig. 2). The total number of the mesh elements in the intermediate grid is 74.6% higher than in the course grid. The fine grid has a total of 3.06 million elements and 5 the prism layers, at the fluid-solid interfaces, with a multiplier of 1.2. The finer grid has a total of 5.23 million elements ($\sim 72\%$ more than in the fine grid), and 7 prim layers with a multiplier of 1.2 in the fluid at interfaces with solid walls.

The obtained values of $\overline{Nu}_{ch}$, based on the results of the present numerical analyses using the different mesh grids are compared in Tables 3 and 4. The results clearly demonstrate that the fine mesh grid is a good compromise between convergence and computation requirement and time. The computation times to complete the numerical analyses in Tables 3 and 4 using the finer grid were ~ 1.7 , and ~ 1.4 times those required to complete the same analyses using the fine grid. To ascertain the accuracy of the present numerical analyses using the fine mesh grid, the results are compared to the reported experimental measurements [11] of the pressure losses in the microchannel in Fig. 2, and the measured temperatures in the Cu substrate along the microchannel length.

2.2. Validation of numerical results

Fig. 4a,b compare the present numerical results of the pressure drop and the water exit temperature in the microchannel, in Figs. 2 and 3, to the experimental values reported by Qu and Mudawar [11]. Fig. 5 compares the calculated temperatures in the Cu substrate to those measured in the experiments [11], using the four thermocouples embedded in the Cu substrate along the microchannel (Fig. 2). The results of the present analyses are indicated in Fig. 4a,b by open symbols, while the reported experimental measurements [11] are indicated by closed symbols. The numerical and the experimental results of the pressure losses and the water exit temperature, at two different values of the heating thermal power, $Q = 20.21$ and 40.45 W, are in excellent agreement (Fig. 4a, b) for a wide range of Re_{in} values (139–1200).

Fig. 5 compares the reported experimental temperature measurements in the Cu substrate [11] to the calculated axial temperature distributions, indicated by the solid lines, for $Re_{in} = 226, 442, 864$ and 890 and $Q = 20.21$ and 40.45 W. The comparisons in Fig. 5, once again, demonstrate excellent agreement between the present numerical results and the reported experimental measurements [11] of the Cu substrate temperature distributions along the

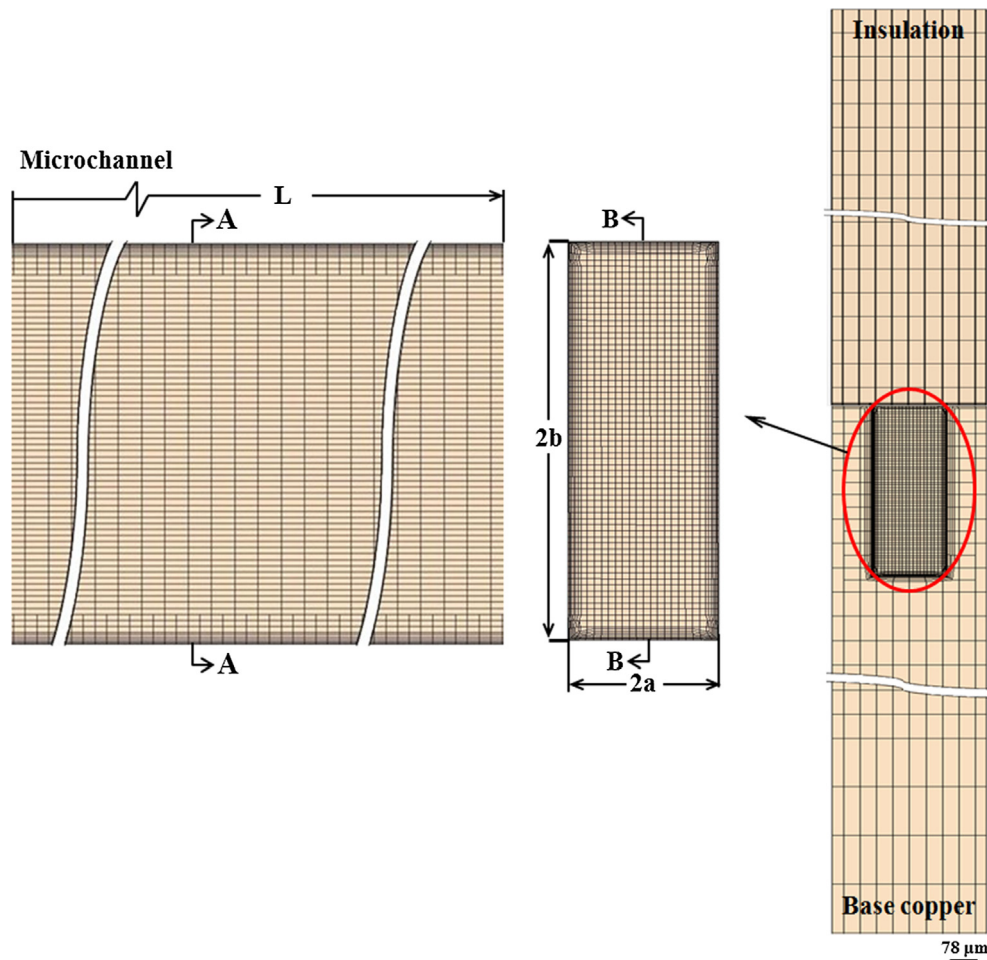


Fig. 3. The fine numerical mesh grid used in present numerical analyses of the microchannel in Fig. 2.

Table 3

Comparison of the numerical results using different mesh grids at $Re_{in} = 139$.

Numerical mesh grid	Mesh elements (million)					No. prism layers/growth multiplier			$\overline{Nu}_{ch}$	
	Total	Fluid		Cu	Insulation	Fluid	Heated solid surface	Insulation	Value	Change (%)
		Prism layer	Bulk							
Coarse	0.716	0.117	0.18	0.33	0.206	3/1.2	3/1.2	3/1.2	5.79	–
Intermediate	1.25	0.45	0.74	0.33	0.206	3/1.2	3/1.2	3/1.2	5.87	~1.36
Fine	3.06	1.473	2.524	0.33	0.206	5/1.2	3/1.2	3/1.2	5.93	~1.01
Finer	5.23	2.92	4.694	0.33	0.206	7/1.2	3/1.2	3/1.2	5.935	~0.085

Table 4

Comparison of numerical results using different mesh grids at $Re_{in} = 890$.

Numerical mesh grid	Mesh elements (million)					No. prism layers/growth multiplier			$\overline{Nu}_{ch}$	
	Total	Fluid		Cu	Insulation	Fluid	Heated solid surface	Insulation	Value	Change (%)
		Prism layers	Bulk							
Coarse	0.716	0.117	0.18	0.33	0.206	3/1.2	3/1.2	3/1.2	7.823	–
Intermediate	1.25	0.45	0.74	0.33	0.206	3/1.2	3/1.2	3/1.2	8.439	~7.3
Fine	3.06	1.473	2.524	0.33	0.206	5/1.2	3/1.2	3/1.2	8.513	~0.87
Finer	5.23	2.92	4.694	0.33	0.206	7/1.2	3/1.2	3/1.2	8.517	~0.05

microchannel (Fig. 2). The comparisons in Figs. 4 and 5 confirm the accuracy of the present numerical results and the fidelity of the implemented methodology using the fine mesh grid (Fig. 3, Tables 3 and 4).

2.3. Methodology of calculating $\overline{Nu}_{ch}$

For the conjugate heat transfer in a microchannel the local wall temperature and heat flow rate, and hence the corresponding local

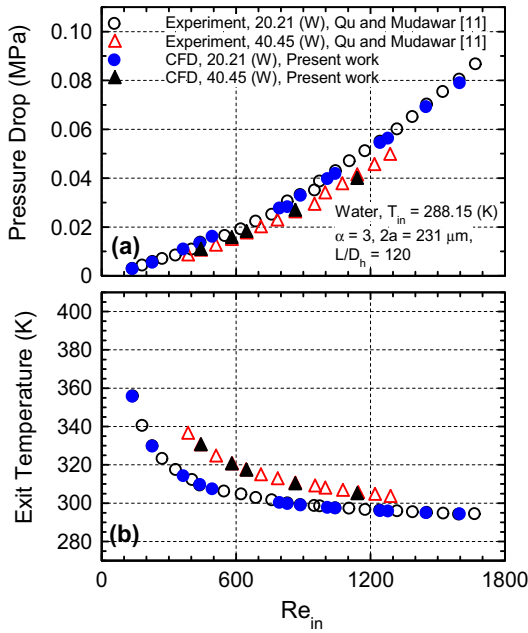


Fig. 4. Comparison of the present numerical results to reported experimental data [11] of the pressure drop and the fluid exit temperature in the investigated microchannel (Figs. 2 and 3).

heat transfer coefficients, would vary azimuthally and axially. At any axial location, z , along the heated microchannel length (e.g., Fig. 2), the local fluid bulk temperature, $T_{f,b}(z)$, and the corresponding values of both the azimuthal heat transfer coefficient, $h(x,y,z)$, and the cross-section average heat transfer coefficient, $h(z)$, are calculated as follows:

$$T_{f,b}(z) = \frac{\int \int \rho u c_p T_f dx dy}{\int \int \rho u c_p dx dy}, \quad (4a)$$

$$h(x,y,z) = \frac{q_w(x,y,z)}{T_w(x,y,z) - T_{f,b}(z)}, \quad \text{and} \quad (4b)$$

$$h(z) = \frac{(\oint h(x,y,z) ds)_{walls}}{2(a+b)} \quad (4c)$$

The local average Nusselt number for the bottom, top, and side walls, $\bar{Nu}_w(z)$, of the microchannel and that of the microchannel, $Nu_{ch}(z)$, are calculated as:

$$\bar{Nu}_w(z) = \frac{\int_{-b}^b h(x,y,z) dy}{2bk_f(z)}, \quad \text{for the side walls} \quad (5a)$$

$$= \frac{\int_{-a}^a h(x,y,z) dx}{2ak_f(z)}, \quad \text{for the top and bottom walls}$$

$$Nu_{ch}(z) = \frac{h(z)D_h}{k_f(z)} \quad (5b)$$

In these expressions, the local thermal conductivity of the fluid, $k_f(z)$, is evaluated at the local fluid bulk temperature in the microchannel, $T_{f,b}(z)$.

The results in Fig. 6a–i show the calculated variations in the wall temperatures, heat flux, and Nusselt number, $Nu_w(z)$, along the side, bottom, and top walls of the microchannel (Fig. 2), at three axial locations ($z/L = 0.25, 0.5, 0.75$). These results are for $Q = 14.01, 20.21$ W and water flow at $Re_{in} = 890$ and $T_{in} = 288.15$ K. The results in Fig. 6 clearly show the large variations in $Nu_w(z)$, which affect the estimates of $\bar{Nu}_w(z)$ (Fig. 7a–c) and of the thermal development length in the microchannel. While determining the values of $Nu_w(z)$ and $\bar{Nu}_w(z)$ is attainable numerically, it is not possible experimentally. In the experiments, the values of the average Nusselt number, $\bar{Nu}_{ch}$, are calculated assuming a constant fluid temperature and a uniform wall heat flux along the microchannel. The fluid temperature is the average of the measured inlet and exit temperatures of the fluid flow in the microchannel, and the wall heat flux is that determined from dividing the total input thermal power, Q , in the experiments by the total area on the inner surface of the microchannel. Therefore, the reported values of $\bar{Nu}_{ch}$ in the experiments would be lower than the determined values, based on the numerical analyses results.

The values of $\bar{Nu}_{ch}$, based on the present numerical analyses are obtained as follows:

$$\bar{Nu}_{ch} = \frac{\int_0^L Nu_{ch}(z) dz}{L} \quad (5c)$$

The expressions in Eqs. (5a)–(5c) indicate that the azimuthal variations of the local heat transfer coefficients affect the axial distribution of the local Nusselt number, $Nu_{ch}(z)$, which in turn affects the estimates of the average Nusselt number, $\bar{Nu}_{ch}$ in the microchannel.

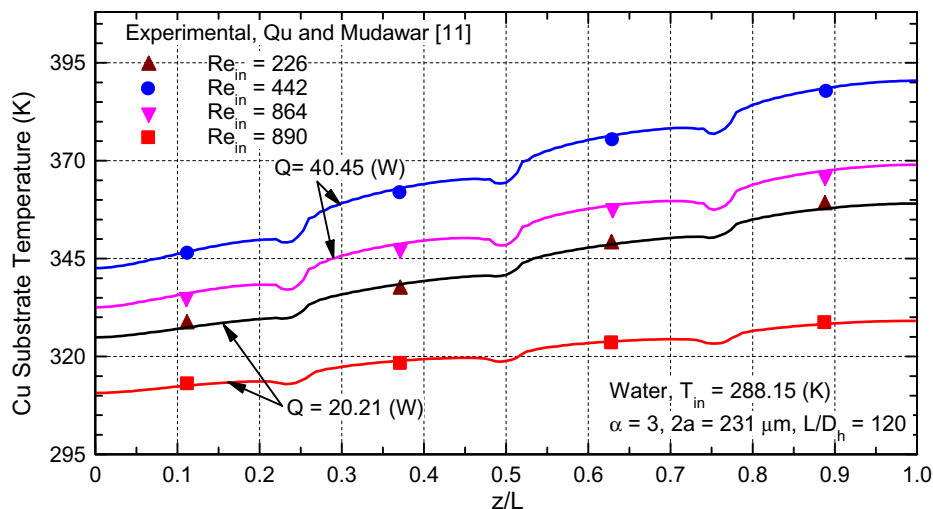


Fig. 5. Comparisons of present numerical results to the reported experimental measurements [11] of the microchannel wall temperatures at different values of Re_{in} and Q .

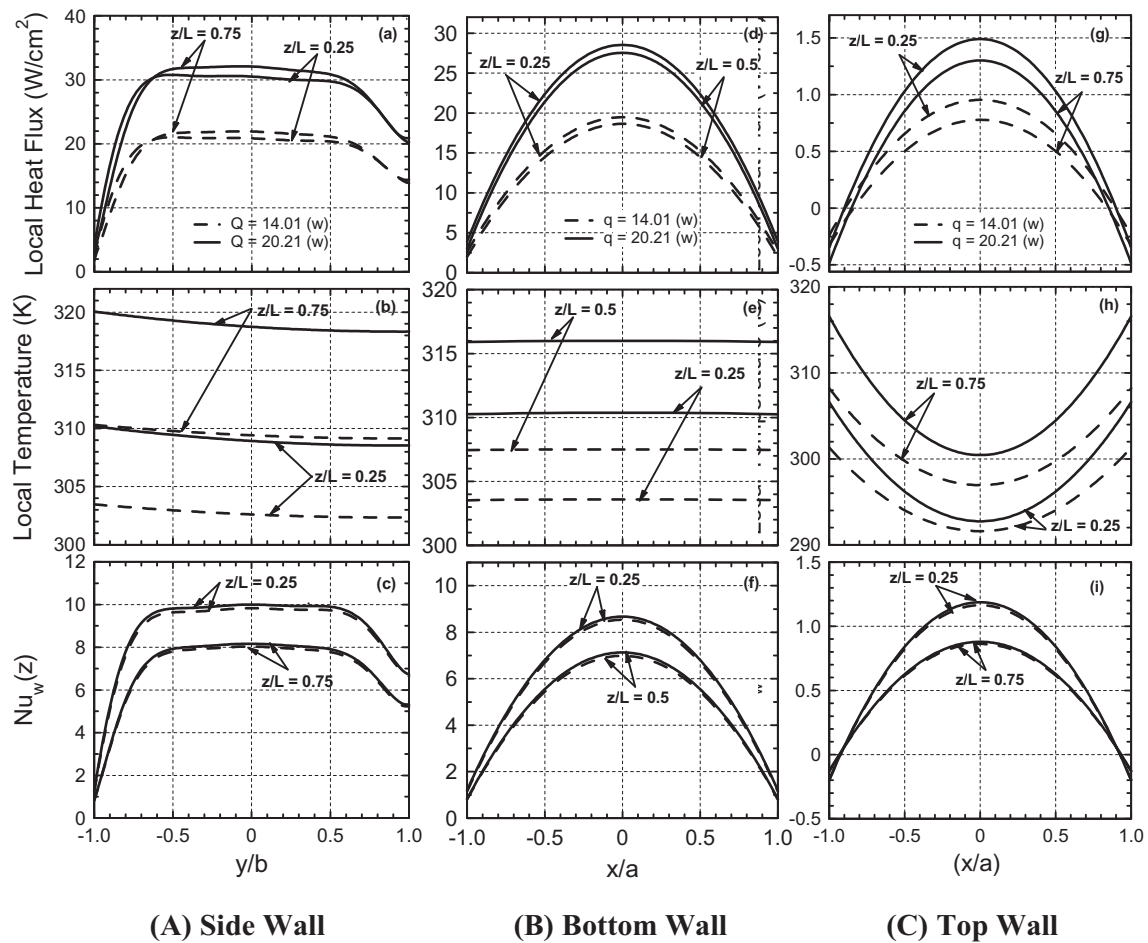


Fig. 6. Calculated values of local heat flux, temperature and average Nusselt number, along the side, bottom, and top walls of the microchannel in Fig. 2, at different axial locations and for $Q = 14.01$ and 20.21 W and $Re_{in} = 890$.

The numerical results delineated in Fig. 6a–c, and g–i, show that the azimuthal variations of the local wall heat fluxes and the corresponding variation in the local Nusselt number are almost uniform in the middle section of the side walls, but drop rapidly near the bottom and the top wall of the microchannel. The top and bottom walls heat fluxes and the corresponding Nusselt numbers have cosine-like azimuthal distributions with peak values at the vertical symmetry plan in the microchannel (Fig. 6b, c, h and i). The local temperatures along the side walls of the microchannel are almost uniform (Fig. 6d), but decrease slightly and gradually with decreased distance from the top insulated wall. The bottom wall temperatures (Fig. 6e) are perfectly uniform, but slightly lower than those of the sides near wall.

For the top wall (Fig. 6f), the azimuthal temperatures have an inverted cosine like distribution, with the lowest value are at the vertical symmetry plan of the microchannel cross-section and the highest values are near the side walls of the microchannel. This inverted temperature distribution indicates that the heat transferred by conduction from the Cu substrate through the vertical Cu walls, to the top insulation, is removed from the top wall mostly by the fluid flowing in the microchannel. The rest of the heat conducted to the top insulation is removed by natural convection of ambient air at its exposed surface (Fig. 2). The azimuthal Nusselt number along the top wall of the microchannel has a cosine-like distribution with the peak occurring at the vertical symmetry plan (Fig. 6j).

Fig. 7a–c compares the calculated axial distributions of the local average Nusselt number, $\overline{Nu}_w(z)$ values for the bottom, side and top

walls of the microchannel (Fig. 2). These distributions demonstrate the developing nature of the fluid flow in the microchannel. The local average values of $\overline{Nu}_w(z)$ for the microchannel cross section give the axial distribution of the local Nusselt number in the microchannel, $Nu_{ch}(z)$. The values of $\overline{Nu}_{ch}$ are then determined from integrating the calculated axial distribution of the local Nusselt number, $Nu_{ch}(z)$, along the microchannel Eq. (5c). The results delineated in Fig. 7a,c for Re_{in} of 139 and 1450, show that the values of $\overline{Nu}_w(z)$ are the highest for the vertical side walls, and the lowest for the top wall of the microchannel. This is because of the differences in the heat flow along the sides, and the bottom and top walls of the microchannel (Fig. 8).

The percentage of the total heat flow through the top wall to the fluid flow is the lowest, while those through the side walls of the microchannel are the highest. The percentage of the heat flow through the bottom of the microchannel in Fig. 2 is higher than that through the top wall, but significantly lower than those through the sides. For the microchannel in Fig. 2, 89% of the applied total thermal power, Q , is delivered to flowing fluid through the side walls, compared to 10% and only 1% through the bottom and the top walls, respectively (Fig. 8). The heat removed by natural convection of ambient air from the exposed surface of the top thermal insulation (Fig. 2) is <0.05%, thus not included in Fig. 8.

The results in Fig. 7a–c show that the values of $\overline{Nu}_w(z)$ increase with increased Q , but not the axial distributions, suggesting that for a give microchannel dimensions (α , D_h) the thermal developed length of $\overline{Nu}_w(z)$ solely depends on Re_{in} . The presented results in

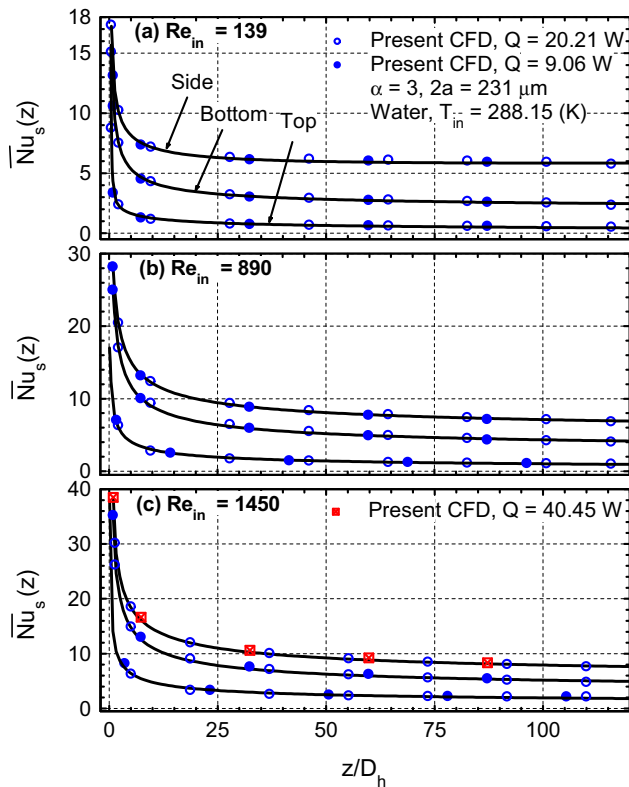


Fig. 7. Calculated average Nusselt numbers, $\overline{Nu}_s(z)$, for the side, bottom and top walls as function of axial location along the microchannel at different values of Re_{in} and Q .

Figs. 7 and 9, of $\overline{Nu}_w(z)$ and $Nu_{ch}(z)$ for water in the microchannel in Fig. 2, are for $Re_{in} = 139$, 890 and 1450, and different values of the total thermal power input, Q . The values of $\overline{Nu}_w(z)$ and $Nu_{ch}(z)$ are highest near the microchannel inlet due to the induced local flow mixing, but decrease gradually with distance from the inlet as the flow gradually develops. They eventually approach asymptotic values near the exit of the microchannel.

The calculated axial distributions of $Nu_{ch}(z)$ for water flow in the heated microchannel in Fig. 2, at different values of Re_{in} and Q (Fig. 9), show that increasing Re_{in} increases the values and raises the axial distribution of $Nu_{ch}(z)$ along the microchannel. This figure also shows that increasing Re_{in} slows down the development of

$Nu_{ch}(z)$ along the microchannel. Note that the length of the microchannel used in the experiments by Qu and Mudawar [11] (Fig. 2) and in the present CFD analyses, whose results are presented in Figs. 4–9, is not long enough for the water flow to become fully developed at the exit. Thus, in order to determine the thermal development length of $Nu_{ch}(z)$, CFD numerical analyses are performed using longer microchannels with different cross-sectional dimensions. The next section presents and discusses the obtained results.

3. Parametric analyses results and discussion

The previous section validated the CFD numerical analyses results, using the fine mesh grid (Tables 3 and 4), for both convergence and accuracy. It also detailed the methodology of accounting for the azimuthal variations in the wall heat fluxes and temperatures used to calculate $Nu_{ch}(z)$ for laminar water flow in the microchannel in Fig. 2 [11]. The results demonstrated the effects of Re_{in} and Q on the values and the axial distribution of $Nu_{ch}(z)$, and the corresponding values of $\overline{Nu}_{ch}$ (Eq. (5c)).

Depending on the value of the flow development length, z_{th} , in the microchannel, $Nu_{ch}(z)$, which increases with increased Re_{in} (Fig. 9), and the actual value of $\overline{Nu}_{ch}$ would be higher than for fully developed flow, Nu_{fd} . Contrasting the numerical values of $\overline{Nu}_{ch}$ to those reported experimentally, based on the averages of the measured wall and fluid bulk temperatures, they would be higher and the difference would depend on the microchannel geometrical parameters, and the values of Re_{in} and Pr_{in} . The performed numerical analyses in the reminder of this paper attempt to quantify the differences between the numerical and experimental $\overline{Nu}_{ch}$ values, for a wide range of geometrical and operating parameters. In addition, the numerical results are used to develop correlations for both $\overline{Nu}_{ch}$ and z_{th} for laminar flows of air and water in microchannels.

3.1. Comparison of numerical and experimental $\overline{Nu}_{ch}$ values

Neglecting the azimuthal variations of the local wall temperatures and heat fluxes, and assuming uniform wall and fluid bulk temperatures in the microchannel are the two main assumptions in the experimental approach for determining $\overline{Nu}_{ch}$ values. Figs. 10 and 11 compare the present numerical values of $\overline{Nu}_{ch}$, which account for the azimuthal variations in the wall temperatures and heat fluxes and the axial variation of the local Nusselt number,

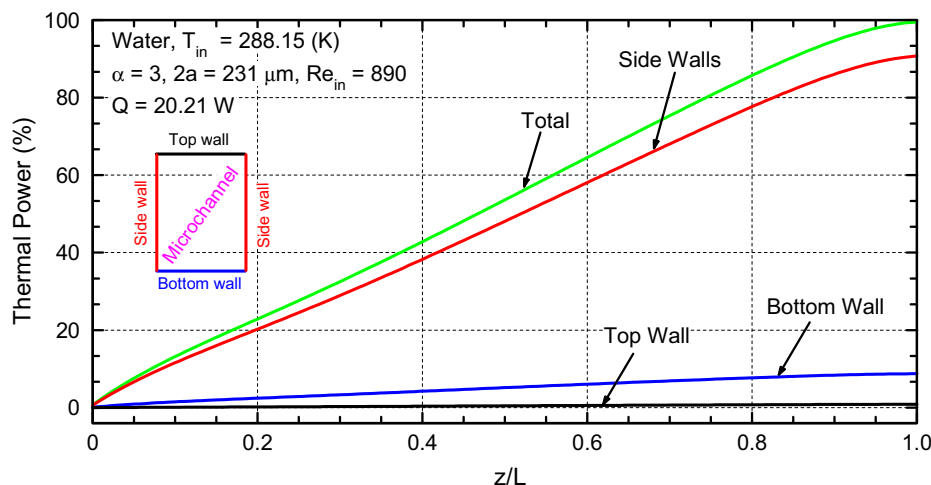


Fig. 8. Axial variations in the heat flow rates through the microchannel walls.

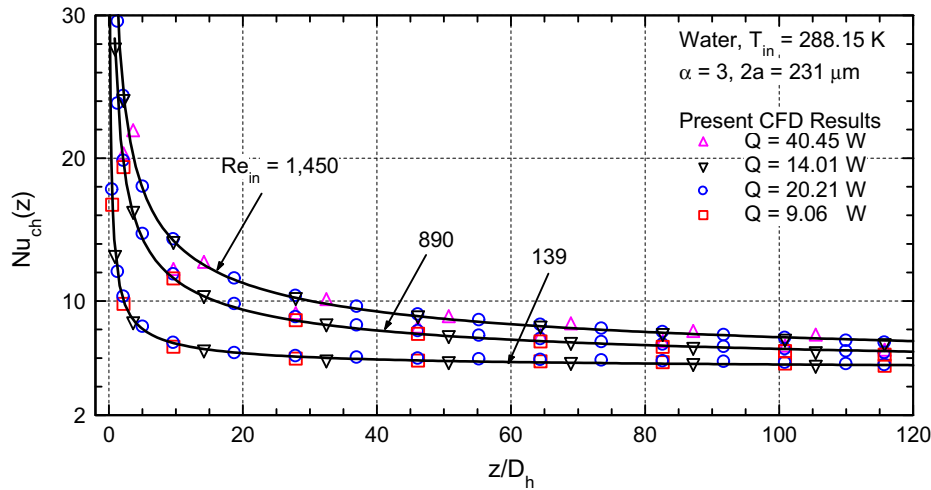


Fig. 9. Effects of Re_{in} and Q on axial distribution of $Nu_{ch}(z)$ along the microchannel in Fig. 2.

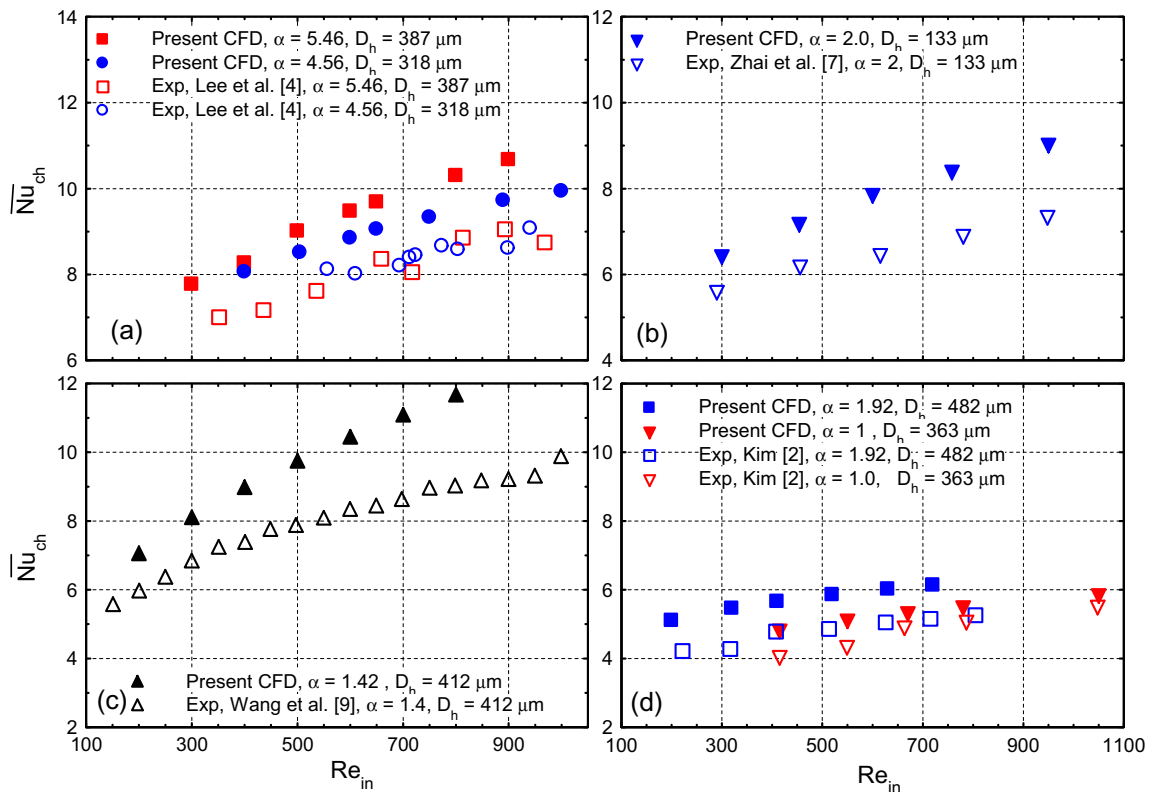


Fig. 10. Comparisons of present numerical and reported experimental values of $\overline{Nu}_{ch}$.

$Nu_{ch}(z)$, to those reported experimentally by other investigators [2,4,7,9].

The comparisons in Figs. 10a–d and 11, show that the reported experimental values of $\overline{Nu}_{ch}$ are consistently lower than those determined numerically in the present work and by other investigators. The difference depends on the geometrical parameters of the microchannels, D_h and α , as well as Re_{in} and Pr_{in} . The results presented in Fig. 11 quantify the differences between the experimental and numerical values of $\overline{Nu}_{ch}$ as a function of Re_{in} . They clearly show that the present numerical values of $\overline{Nu}_{ch}$ are in excellent agreement with those reported by other investigator [8,12,26], and that the reported experimental values [2,4,7,9] are up to 20%

lower than the numerical values. Thus, experimentally reported values and correlations (Table 1 and Fig. 1) would underestimate $\overline{Nu}_{ch}$ estimates for practical applications, including the design and performance prediction of the microchannels heat sinks.

The present numerical results also show that for a uniform inlet flow velocity, the values of $Nu_{ch}(z)$ are not uniform along the microchannel length, but higher at the entrance and decrease gradually with distance into the microchannel (Figs. 9 and 12). They eventually reach the same asymptotic value, Nu_{fd} , for fully developed flow, which unlike the thermal development lengths, l_{th} , is independent on Re_{in} . The results presented in Figs. 12 and 13 show that Nu_{fd} , near the exit of the microchannel, is independent of Pr_{in}

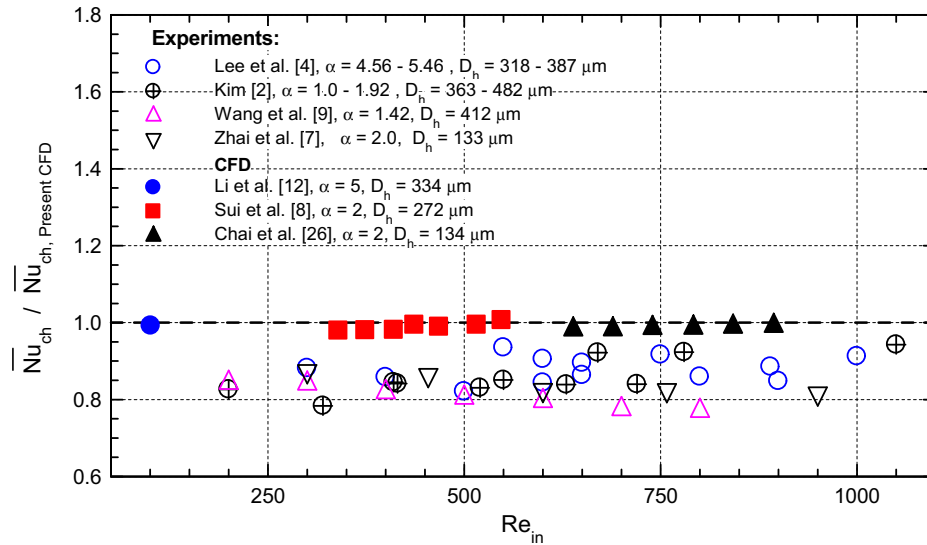


Fig. 11. Comparison of the present numerical analyses results to those reported by others and to reported experimental values of $\overline{Nu}_{ch}$ for water flow in microchannels.

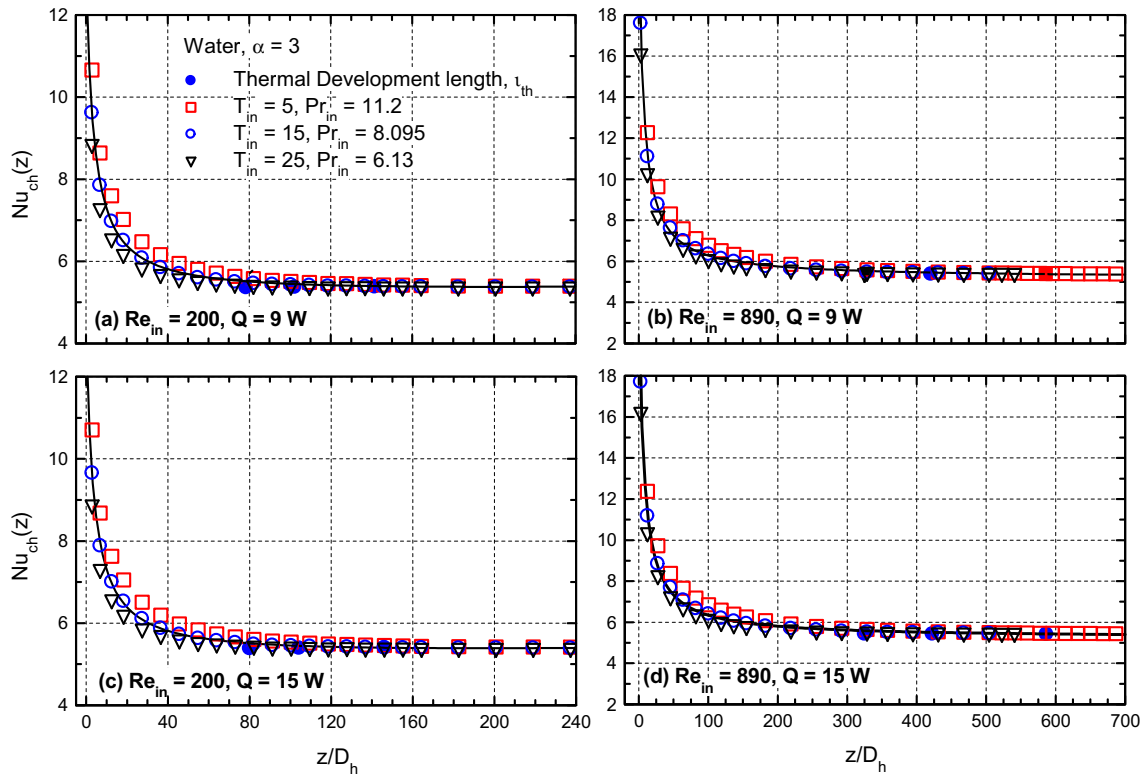


Fig. 12. Calculated values of $Nu_{ch}(z)$ for water flow at different values of Re_{in} and Pr_{in} in a heated microchannel with $\alpha = 3$.

and Re_{in} , but increases with increased α , (Fig. 14). However, determining the $\overline{Nu}_{ch}$ values requires accurate determination of the thermal development length, l_{th} . The present results show that the value of l_{th} depends on Re_{in} , Pr_{in} , and α . It increases with increased Re_{in} and/or decreased α (Figs. 12, 13, 15, 16).

3.2. Development length for $Nu_{ch}(z)$

Determining the development length, l_{th} , for $Nu_{ch}(z)$ is not practically attainable experimentally, but could be obtained from the numerical analyses results of the conjugate heat transfer in

microchannels. The results of the present numerical analyses, which account for the temperature dependent properties of the flowing fluid in the microchannel, are used to determine the values of l_{th} in sufficiently long microchannels, for wide ranges of geometrical parameters and Re_{in} values. Li et al. [12] have shown the importance of accounting for the changes in the fluid properties with temperature when calculating the pressure losses and the convective heat transfer in microchannels. Unfortunately, most reported correlations for l_{th} [13–15] are based on numerical results that assume constant fluid properties.

In this section, comprehensive numerical analyses are performed of the conjugate heat transfer for laminar flow of air and

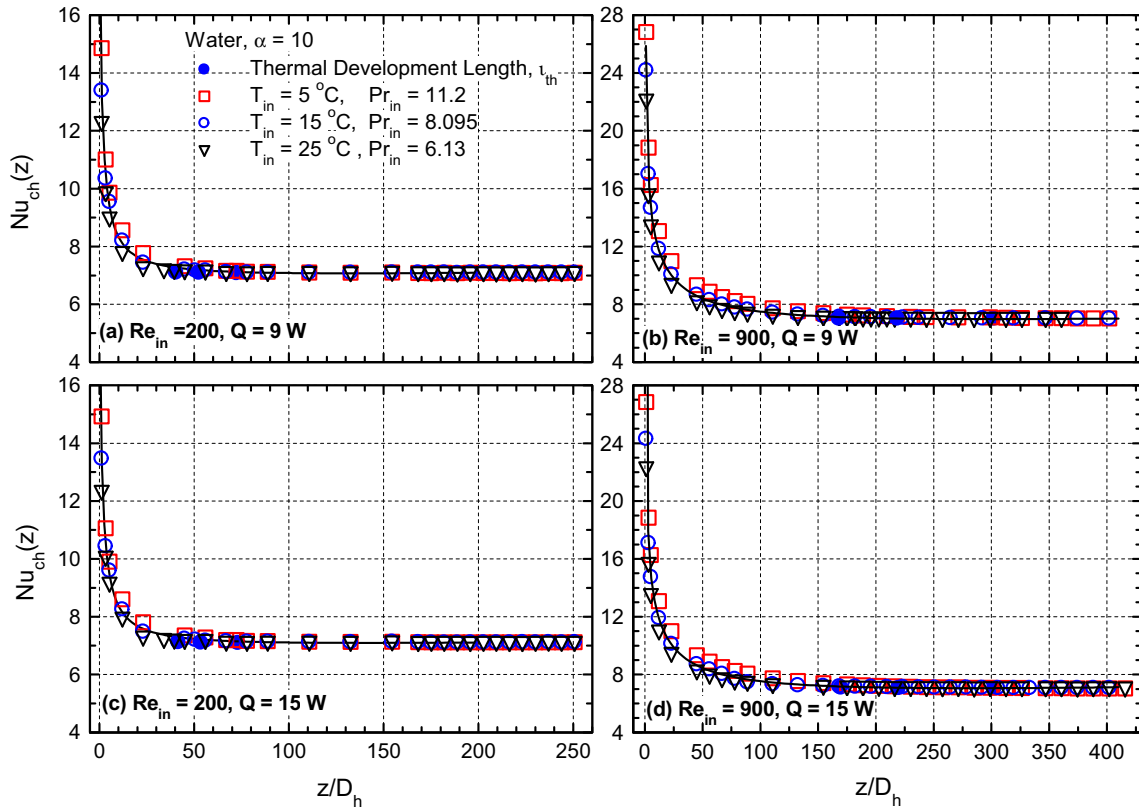


Fig. 13. Calculated $Nu_{ch}(z)$ versus the dimensional distance for the microchannel entrance for different values of Re_{in} and Pr_{in} and for $\alpha = 10$.

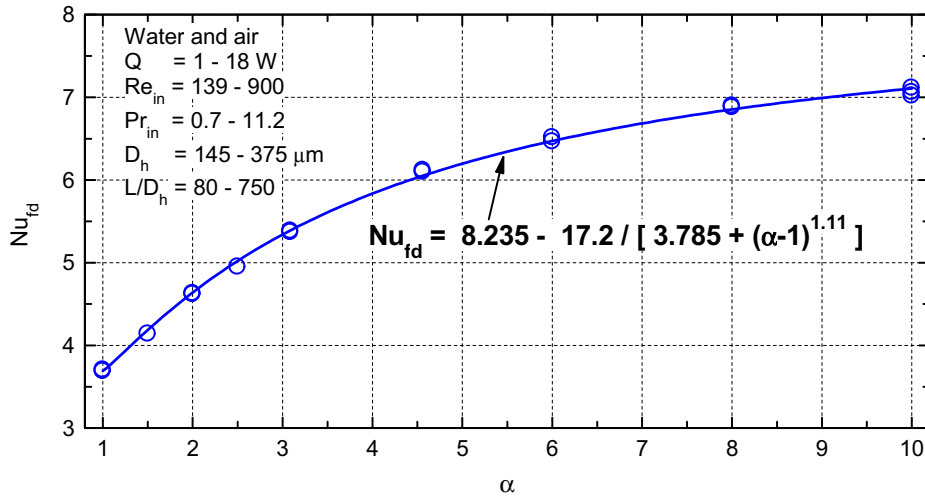


Fig. 14. Calculated values of Nu_{fd} and the developed correlation, solely as a function of, α .

water in long microchannels, with $L/D_h = 130\text{--}750$, to determine the values of l_{th} for wide ranges of parameters. These include $\alpha = 1\text{--}10$, $D_h = 145\text{--}375\ \mu\text{m}$, $Re_{in} = 130\text{--}900$, $Pr_{in} = 0.7\text{--}11.2$, and $Q = 1\text{--}18\ \text{W}$. The results in Figs. 12 and 13 demonstrate the development of $Nu_{ch}(z)$ along the microchannel length, for different values of Pr_{in} and $\alpha = 3$ and 10, respectively. The development length, l_{th} , indicated in these figures by solid symbols, is measured from the entrance of the microchannel to when $Nu_{ch}(z)$ is within 1% of its fully developed value, Nu_{fd} .

The results in Figs. 12 and 13, show that l_{th} depends not only on Re_{in} and Pr_{in} , but also on Q . It is inferred from the results in these

figures that increasing Pr_{in} increases l_{th} and raises the local Nusselt number, $Nu_{ch}(z)$, near the microchannel entrance. Moreover, $Nu_{ch}(z)$ approaches the same fully developed value, Nu_{fd} , regardless of the values of either Pr_{in} , Re_{in} . However, Nu_{fd} increases as the aspect ratio of the microchannel, α , increases from 3 to 10 (Figs. 12–14). The results on Fig. 14 illustrate that Nu_{fd} increases with increasing, α , and is independent of the values of Q , Re_{in} , and Pr_{in} . The values of Nu_{fd} in Figs. 13 and 14, which are for, $D_h = 145\text{--}375\ \mu\text{m}$, and $L/D_e = 130\text{--}750$, are correlated, as:

$$Nu_{fd} = 8.235 - \left(17.2 / \left[3.785 + (\alpha - 1)^{1.11} \right] \right) \quad (6)$$

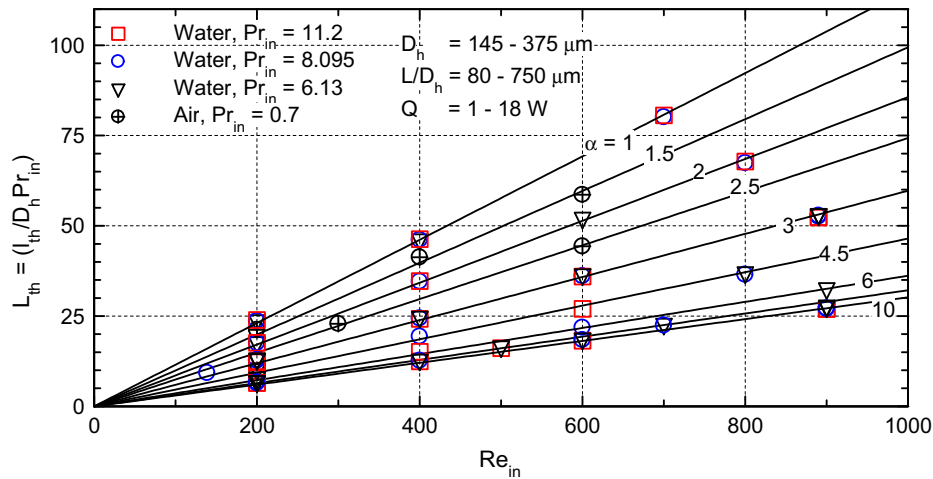


Fig. 15. Calculated dependence of the dimensionless thermal development length of $Nu_{ch}(z)$ in microchannels on Re_{in} and α .

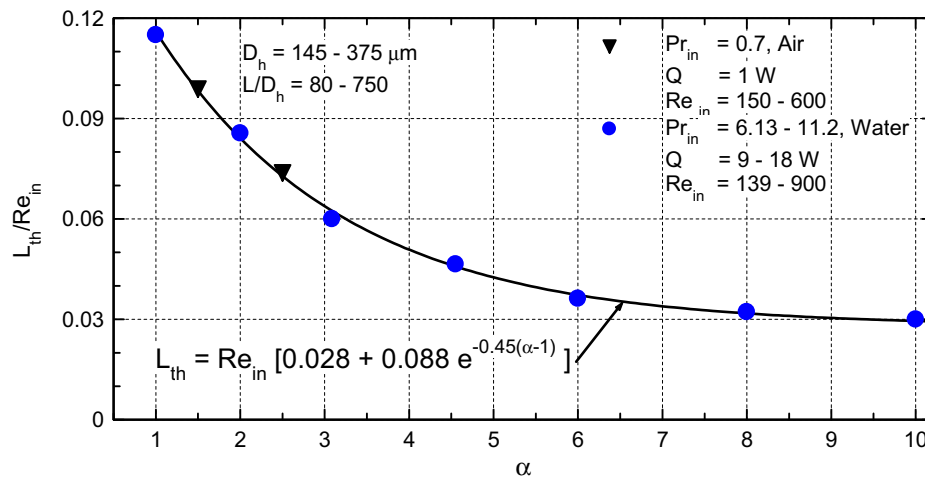


Fig. 16. Developed correlation of dimensionless thermal development length in microchannels.

This correlation is in excellent agreement with the present numerical results, showing that Nu_{fd} is independent of Re_{in} and Pr_{in} , but increases monotonically with increased α . It increases from ~ 3.7 for square microchannels ($\alpha = 1$) to ~ 7.1 for rectangular microchannels with $\alpha = 10$ (Fig. 14).

The present numerical results for laminar flows of water and air in microchannels are also used to develop a correlation for the thermal development length, l_{th} , of $Nu_{ch}(z)$, for a wide range of geometrical and operation parameters (Fig. 15). This figure plots the dimensionless thermal development length, $L_{th} = (l_{th}/D_h Pr_{in})$, versus Re_{in} for microchannel aspect ratios, $\alpha = 1.0$ to 10 . For a given α , the dimensionless development length increases linearly with increased Re_{in} . Also, for a given Re_{in} , the dimensionless development length of $Nu_{ch}(z)$ decreases with increased α , and is highest for the square microchannels. The obtained values of the dimensionless thermal development length in Fig. 15 are correlated in Fig. 16, as:

$$L_{th} = Re_{in} (0.028 + 0.088 e^{-0.45(\alpha-1)}). \quad (7a)$$

This correlation is in excellent agreement with the present numerical results. It clearly shows that the thermal development length, l_{th} , of $Nu_{ch}(z)$, depends not only on α and Re_{in} , but also D_h and Pr_{in} ; thus:

$$l_{th} = (0.028 + 0.088 e^{-0.45(\alpha-1)}) Re_{in} Pr_{in} D_h. \quad (7b)$$

3.3. $Nu_{ch}(z)$ correlation

In this section, the present numerical analyses results of $Nu_{ch}(z)$ for laminar flows of water and air in microchannels are correlated (Fig. 17), as:

$$Nu_{ch}(z) = Nu_{fd} [1.0 + (0.0134/z^{*0.631})], \quad (8)$$

where

$$z^* = [(z/D_h)(\alpha^{0.583}/Re_{in} Pr_{in})]^{1.2}. \quad (9)$$

As delineated in Fig. 17, the correlation in Eq. (8) is in excellent agreement with the present CFD numerical results for laminar flows of water and air in microchannels with $\alpha = 1-10$, $D_h = 145-375$, and $L/D_h = 130-750$, at $Re_{in} = 130-900$, $Pr_{in} = 0.7-11.2$, and $Q = 1$ and $9-18$ W. This correlation is also in excellent agreement with the reported numerical results by other investigators [12,27], using similar numerical procedures and accounting for the changes in the fluid properties with temperature (Fig. 18).

3.4. $\overline{Nu}_{ch}$ correlation

The values of the microchannel average Nusselt number, $\overline{Nu}_{ch}$, for laminar flows of water and air are calculated from the integration of the obtained expression of $Nu_{ch}(z)$, given in Eq. (8), over the

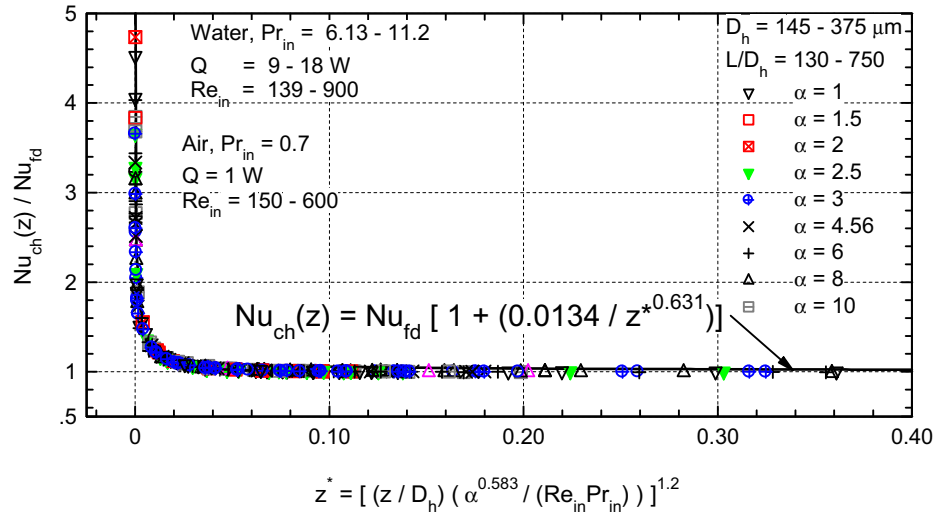


Fig. 17. Comparison of developed $Nu_{ch}(z)$ correlation for microchannels with the present CFD numerical results for laminar flows in microchannels.

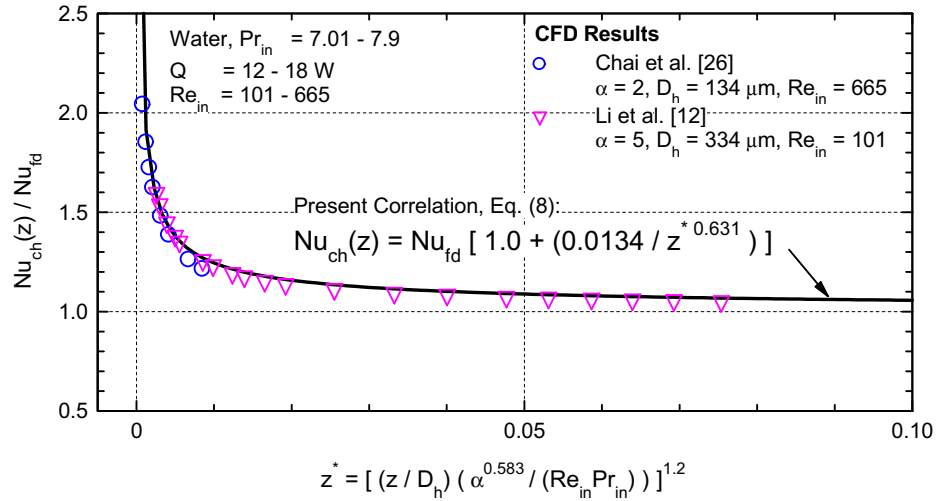


Fig. 18. Comparison of the developed $Nu_{ch}(z)$ correlation, (Eq. (8)), to the reported numerical results by other investigators [12,27].

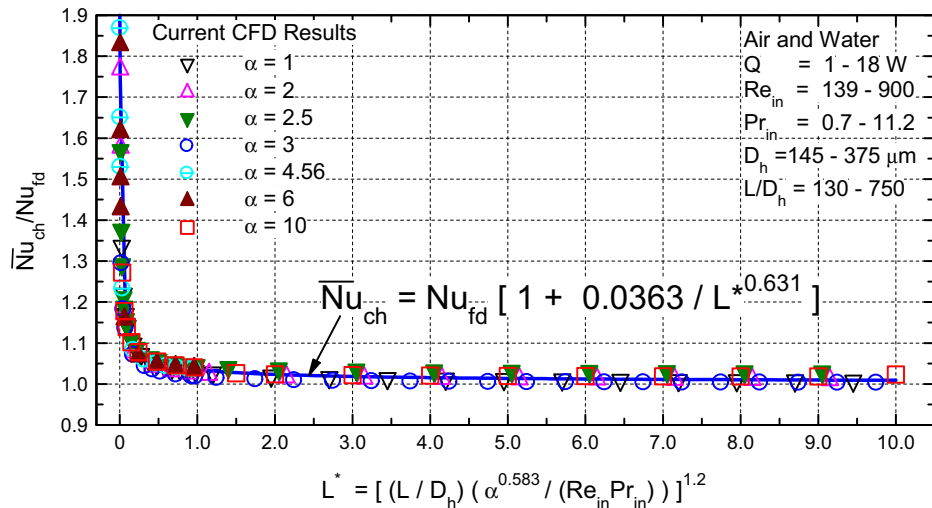


Fig. 19. Comparison of developed $\overline{Nu}_{ch}$ correlation, Eq. (10), with present numerical results.

microchannel total length (Eq. (5c)). The obtained values are correlated in terms of Nu_{fd} and a dimensionless microchannel length, L^* , as:

$$\overline{Nu}_{ch} = Nu_{fd} \left[1 + 0.0363 / L^{*0.631} \right], \quad (10)$$

where

$$L^* = \left[(L/D_h) (\alpha^{0.583} / Re_{in} Pr_{in}) \right]^{1.2}. \quad (11)$$

The comparisons in Fig. 19 also show excellent agreement of the present numerical results for air and water flows in microchannels with the developed continuous correlation of $\overline{Nu}_{ch}$ (Eq. (10)). The comparison in Fig. 20 also demonstrate that the present $\overline{Nu}_{ch}$

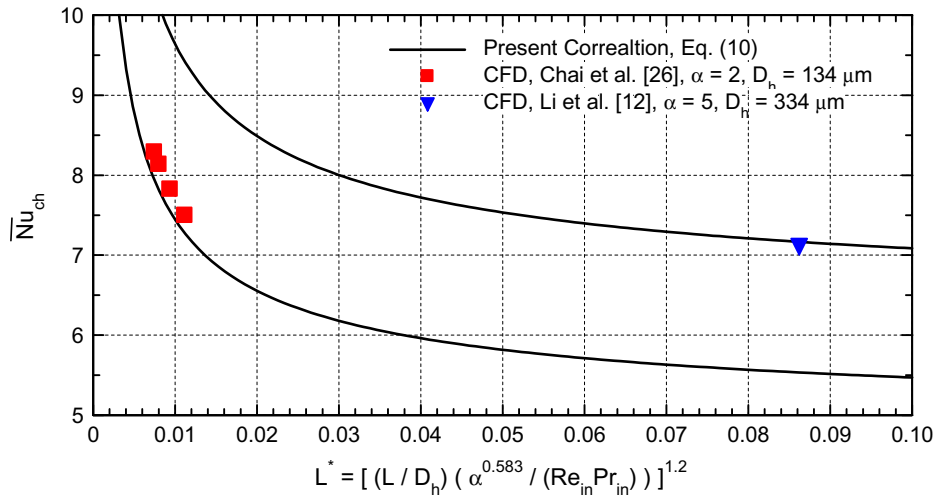


Fig. 20. Comparison present $\overline{Nu}_{ch}$ correlation (Eq. (10)) with numerical results by others [12,27].

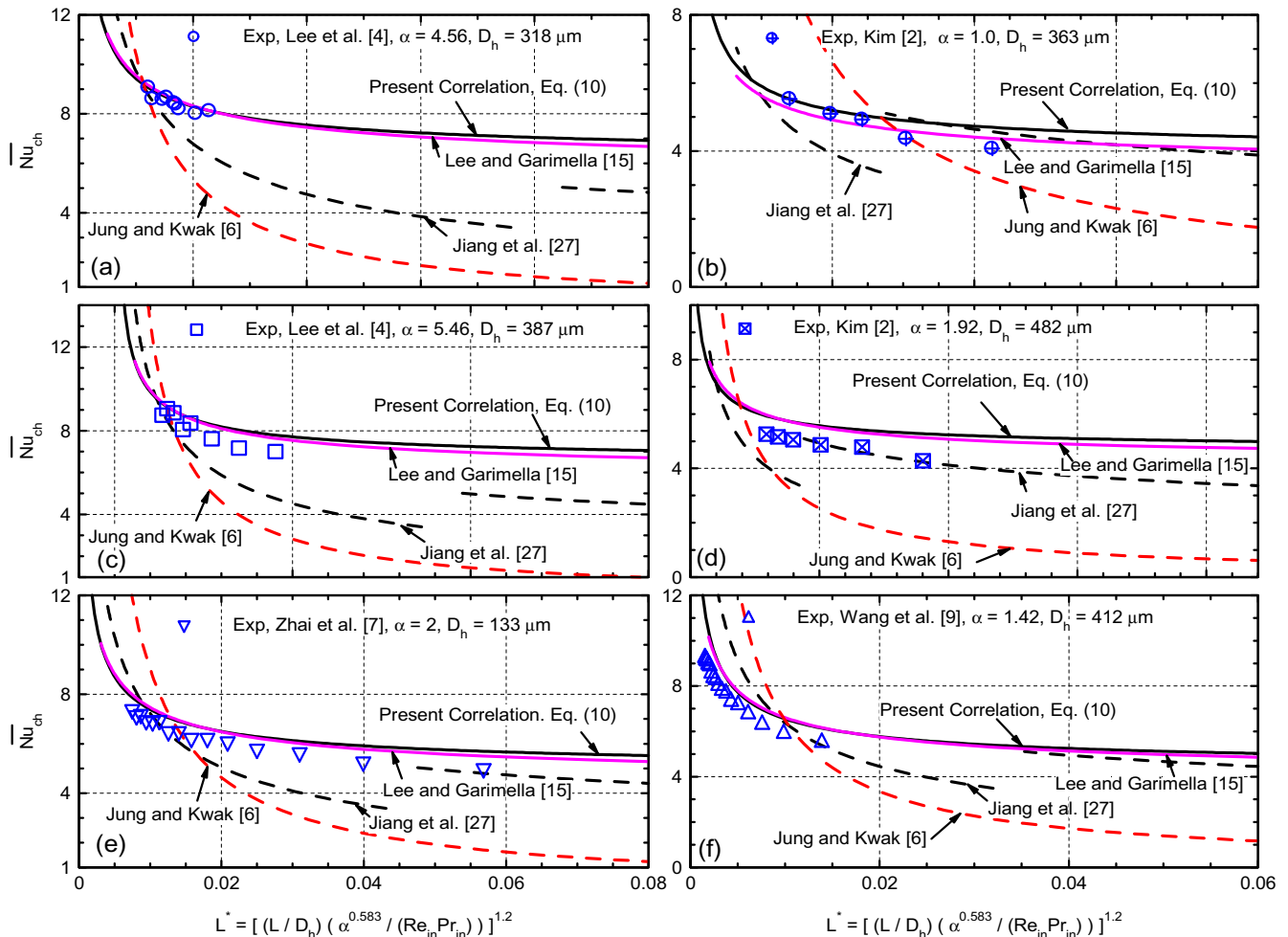


Fig. 21. Comparison of present $\overline{Nu}_{ch}$ correlation with reported experimental data and correlations by other investigators.

correlation (Eq. (10)) is in good agreement with the reported numerical results by other investigators [12,27] for water flow in microchannels with $\alpha = 2$ and 5 and $D_h = 134$ and $334 \mu\text{m}$. The present correlation of $\overline{Nu}_{ch}$ (Eq. (10)) is compared in Fig. 21a–f with the reported experimental data and $\overline{Nu}_{ch}$ correlations by others (Fig. 1 and Table 1). These figures show that the present correlation (Eq. (10)) is consistently higher than that reported by Lee and Garimella, also based on numerical analyses [15]. Unlike the present correlation, that of Lee and Garimella is based on assuming a uniform wall heat flux and isothermal fluid temperature. Both correlations are 10–20 higher than the experimental results reported by various investigators for water flows in microchannels with $\alpha = 1.0, 1.42, 1.92, 2, 4.56$ and 5.46 and $D_h = 133\text{--}482 \mu\text{m}$ (Fig. 21a–f).

As indicated earlier, the reported experimental values of $\overline{Nu}_{ch}$ are based on the measured average wall and fluid temperatures in the experiments, and neglect the azimuthal variations in the wall heat flux and temperature. They also assume constant fluid properties, which are determined either at the inlet or bulk temperature in the microchannel. The correlation reported by Jiang et al. [27] is discontinuous and significantly under predict $\overline{Nu}_{ch}$, and that by Jung and Kwak [7] is continuous by predicts the lowest values of $\overline{Nu}_{ch}$ (Fig. 21).

4. Summary and conclusions

Extensive numerical analyses of the 3-D conjugate heat transfer of laminar flows of water and air in microchannels are performed. They account for the azimuthal variations of the wall heat flux and temperature, and the temperature dependent fluid properties. The convergence of the results using the implemented methodology and the numerical mesh refinement is confirmed, and the accuracy is demonstrated by the excellent comparison with the reported experimental measurements of the pressure drop, exit water temperature, and the axial wall temperatures along the microchannel in the experiments of Qu and Mudawar [11].

Subsequently, parametric numerical analyses are carried out to generate a large database for correlating the local Nusselt number, $Nu_{ch}(z)$, and the flow thermal development length, z_{th} , in the microchannels. The obtained axial distributions of $Nu_{ch}(z)$ are integrated over the microchannel length to obtain the values of the $\overline{Nu}_{ch}$. The wide range of parameters investigated in the present numerical analyses include $\alpha = 1\text{--}10$, $D_h = 145\text{--}375 \mu\text{m}$, $L/D_h = 130\text{--}750$, $Q = 1\text{--}18 \text{ W}$, and $Re_{in} = 130\text{--}900$, $Pr_{in} = 0.7\text{--}11.2$.

The present numerical results are used to develop continuous correlations for $Nu_{ch}(z)$ and $\overline{Nu}_{ch}$ as well as for the thermal development length of $Nu_{ch}(z)$, z_{th} , in microchannels in terms of the geometrical parameters (α , D_h , L/D_h) and the operating conditions (Re_{in} , Pr_{in} and Q). The developed correlations are as follows:

(1) Nusselt Numbers in microchannels:

$$Nu_{ch}(z) = Nu_{fd} [1.0 + 0.0134/z^{0.631}],$$

and,

$$Nu_{fd} = 8.235 - \left(17.2 / [3.785 + (\alpha - 1)^{1.11}] \right)$$

(2) Thermal Development length of $Nu_{ch}(z)$:

$$z_{th} = Re_{in} Pr_{in} D_h [0.028 + 0.088 e^{-0.45(\alpha-1)}].$$

(3) Microchannel Average Nusselt Number:

$$\overline{Nu}_{ch} = Nu_{fd} [1 + 0.0363/L^{*0.631}], \text{ where } L^* = [(L/D_h)(\alpha^{0.583}/Re_{in}Pr_{in})]^{1.2}$$

The $\overline{Nu}_{ch}$ correlation is in good agreement with the numerical results reported by others. The experimental values of $\overline{Nu}_{ch}$ are consistently 10–20% lower than the numerical values. This is because the experimental values are based on the average wall and fluid temperatures in the experiments, and constant fluid properties. Thus, unlike the present numerical analyses, they do not account for the azimuthal variations in the wall temperature and heat flux, the flow development at the microchannel entrance, and the change in the fluid properties with temperature.

In conclusion, while it is possible to determine the values of $Nu_{ch}(z)$ and z_{th} using numerical analyses, these quantities are practically impossible to determine experimentally. The reported correlations of $Nu_{ch}(z)$, z_{th} , and $\overline{Nu}_{ch}$ are a valuable contribution to the design and the performance prediction of microchannel equipment for a wide range of industrial and engineering applications. These include heat sinks for cooling electronics, high power computer chips, and CPUs, compact heat exchangers, solar energy, and thermal management of spacecraft and laser diodes and sensors.

Conflict of interest

The authors declared that there is no conflict of interest.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ijheatmasstransfer.2018.12.077>.

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